Siddharth Suman Pandey

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Machine Learning & GenAI Engineer | Python | LLMs | NLP | AI Applications

Machine Learning and GenAI Engineer with 3+ years of experience building Python-based applications. Experienced in machine learning, NLP, LLM integration, and recommendation systems. Skilled in developing production APIs, optimizing data pipelines, and deploying AI solutions using Azure.

WORK EXPERIENCE

Entelika **12/2023 - Present**
Full Stack AI Software Engineer • Full-time **Mysore**
Team Lead

- Led development of TieOpp, a career platform serving students and enterprises, delivering interview coaching, personality assessment, and personalized opportunity recommendations.
- Developed Python services and FastAPI endpoints for AI features, connecting model outputs, external LLM APIs, and application data.
- Implemented an LLM-powered interview coaching system using Azure OpenAI GPT-4o-mini to analyze speech transcripts and generate contextual feedback on fluency, grammar, pronunciation, and interview responses.
- Built candidate evaluation workflows that combine assessment responses and AI-generated feedback to help candidates understand their strengths and areas for improvement.
- Developed a semantic recommendation system using Sentence-BERT embeddings and similarity matching, then optimized retrieval with index-based matching to reduce processing overhead and improve response times.
- Designed and optimized SQL schemas, queries, and stored procedures to support AI-driven applications, efficient data retrieval, and high-volume processing workloads.
- Improved production AI processing reliability by implementing asynchronous task queuing with Redis to manage concurrent workloads and reduce bottlenecks in LLM-powered application workflows.
- Optimized data transformation workflows by replacing 1,207+ SQL scripts with four Python scripts using reusable logic, dynamic filtering, and parallel processing, achieving approximately 7× faster execution.
- Delivered data engineering and modernization solutions for Cabot Corporation and ENOWA (NEOM), improving pipeline efficiency and maintainability.

RenewBuy **11/2022 - 10/2023**
Full Stack AI Developer Trainee • Full-time **Gurgaon**
Intern

- Developed a machine learning-based sales prediction system achieving 98.4% accuracy, applying predictive analytics to identify sales patterns and support data-driven business decisions.
- Integrated predictive models into REST APIs using Django REST Framework and developed Angular dashboards for accessing model outputs and analytical insights.
- Created interactive analytics visualizations to analyze historical sales data, identify trends, and communicate prediction results.
- Engineered an ML-based vehicle Registration Certificate (RC) recognition system achieving approximately 94% accuracy using OCR, image preprocessing, and computer vision techniques for document information extraction and validation.

EDUCATION

B.Tech – Computer Science & Engineering
Abdul Kalam Technical University • GPA: 8.2 08/2019 - 08/2023

12th Science
Shiv Jyoti Convent Senior Secondary School 03/2018

AWARDS & SCHOLARSHIPS

Smart India Hackathon – 3rd Position
Uttarakhand Govt.

Hacknovate 2.0 – 3rd Position
Jamia Milia Islamia

PROJECTS

Lane Line Detection using Machine Learning
Research Project | IEEE Publication

Developed a real-time lane detection system using computer vision techniques to identify road lane boundaries under challenging driving conditions. Applied Canny Edge Detection and the Hough Transform to process video frames and detect lane markings, supporting improved lane visibility and road safety.

Technologies: Python, OpenCV, NumPy, Computer Vision, Image Processing, Canny Edge Detection, Hough Transform, Real-Time Video Processing.

Research Paper: https://ieeexplore.ieee.org/document/10064598

AI Video Coach
TieOpp

Built an LLM-powered interview coaching system using Azure OpenAI GPT-4o-mini. Built a pipeline to process recorded interview responses, generate speech transcripts, and analyze answers in context to provide personalised feedback on grammar, pronunciation, fluency, repetition, and communication skills. Integrated video storage, transcription, LLM inference, and structured feedback persistence using Python, FastAPI, Azure Blob Storage, and SQL. Implemented Redis based task queuing to improve workload management and production scalability.

Technologies: Python, FastAPI, Azure OpenAI, GPT-4o-mini, LLMs, NLP, Speech-to-Text, Redis, Azure Blob Storage, SQL, REST APIs.

SKILLS

Programming: Python, SQL

Machine Learning & NLP: Machine Learning, Predictive Analytics, Recommendation Systems, Semantic Embeddings, Sentence-BERT, Natural Language Processing (NLP), OCR, Computer Vision, Image Processing

Generative AI: Large Language Models (LLMs), Azure OpenAI, GPT-4o-mini, Prompt Engineering, LLM API Integration, Contextual Text Analysis

Backend & APIs: FastAPI, Flask, Django REST Framework, REST APIs

Data Engineering: Azure Synapse Analytics, Data Pipelines, SQL Optimization, Parallel Processing, Data Transformation

Databases & Infrastructure: MySQL, Redis, Azure Blob Storage, Azure App Services

Frontend: Angular, React

Development Tools: Git, Azure DevOps, Postman, CI/CD Pipelines